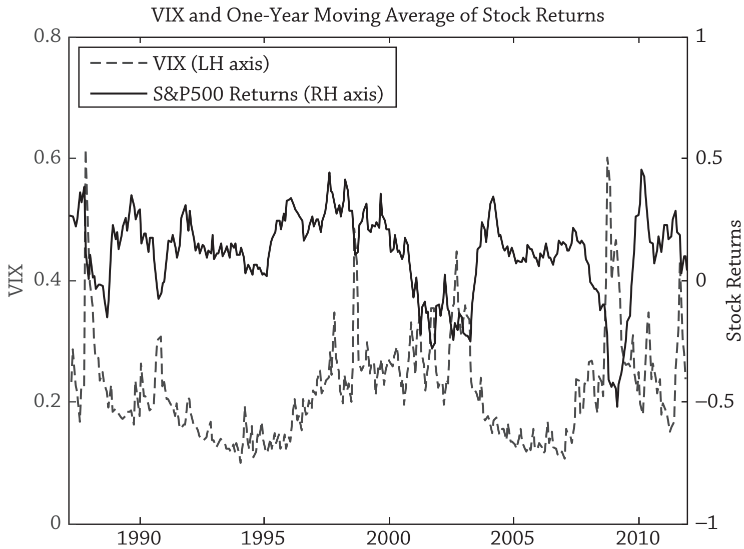

Figure 7.3

Figure 7.3 plots the VIX index (left-hand side axis) in the dashed line and a one-year moving average of stock returns (on the right-hand side axis) in the solid line. Volatility tends to exhibit periods of calm, punctuated by periods of turbulence. Figure 7.3 shows spikes in volatility corresponding to the 1987 stock market crash, the early 1990s recession, the Russian default crisis in 1998, the terrorist attacks and ensuing recession in 2001, and a large spike in 2008 corresponding to the failure of Lehman Brothers. In all of these episodes, stock returns move in the opposite direction from volatility, as shown during the financial crisis.

The losses when volatility spikes to high levels can be quite severe. Partitioning the sample into high and low periods of volatility changes gives us an average return for large stocks of –4.6% during volatile times and 24.9% during stable times. This compares to an overall mean of 11.3% (see Table 7.2). Investors allergic to volatility could increase their holdings of bonds, but bonds do not always pay off during highly volatile periods—as the low correlation of 0.12 between VIX changes and bond returns in the table above shows.

Stocks are not the only assets to do badly when volatility increases. Volatility is negatively linked to the returns of many assets and strategies. Currency strategies fare especially poorly in times of high volatility.[10] We shall see later that many assets or strategies implicitly have large exposure to volatility risk. In particular, hedge funds, in aggregate, sell volatility (see chapter 17).

Investors who dislike volatility risk can buy volatility protection (e.g., by buying put options). However, some investors can afford to take on volatility risk by

[10] See Bhansali (2007) and Menkhoff et al. (2012a).